

1 Emergence of Lorentz Symmetry from Stew Structure

Premises. Let t_P be the fundamental gate tick and L the correlation length. A body moving at speed v creates local disturbances at rate

$$f_{\text{ripple}}(v) = \frac{v}{L}. \quad (1)$$

The stew can process at most one quantum per tick: $f_{\text{max}} = 1/t_P$. Planck calibration $L \equiv c t_P$ gives

$$\beta := \frac{v}{c} = \frac{f_{\text{ripple}}}{f_{\text{max}}}, \quad 0 \leq \beta < 1, \quad (2)$$

hence $v_{\text{max}} = L f_{\text{max}} = c$ is a *structural* limit, not a postulate.

Gate widening and time dilation. As load grows ($\beta \uparrow$) the stew widens its effective quantization gate. We model the per-tick delay as

$$\Delta t_q(v) = t_P [1 + p_{\text{emit}}(\beta)], \quad (3)$$

with $p_{\text{emit}}(0) = 0$ and $p'_{\text{emit}}(\beta) > 0$. To reproduce relativistic time dilation we require

$$\frac{\Delta t_q(v)}{t_P} = \gamma(\beta) = \frac{1}{\sqrt{1 - \beta^2}}. \quad (4)$$

A minimal consistent choice is therefore

$$p_{\text{emit}}(\beta) = \gamma(\beta) - 1. \quad (5)$$

Modeling note (common pitfall). If one instead postulates $p_{\text{emit}}(\beta) = (1 - \beta^2)^{-1} - 1 = \gamma^2 - 1$, then (??) yields $\Delta t_q = t_P \gamma^2$, which *over-dilates* time by an extra factor γ . Equation (??) fixes this.

Summary. With $L = c t_P$ and $p_{\text{emit}} = \gamma - 1$, the invariant speed c and the Lorentz factor γ emerge from *finite throughput* and *gate adaptation*:

$$v_{\text{max}} = c, \quad \Delta t_q(v) = t_P \gamma, \quad (\text{and two-way light speed is isotropic under coarse graining}).$$